A Magnetic-reference fraction

MatterGen base
MatterGen tuned
Con-CDVAE base
Con-CDVAE tuned
Crys-JEPA base
Crys-JEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned

0.0 0.2 0.4 0.6 0.8 1.0

Fraction closer to magnetic than NM reference

AFM FM FiM

B Magnetic-reference contrast

MatterGen base
MatterGen tuned
Con-CDVAE base
Con-CDVAE tuned
Crys-JEPA base
Crys-JEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned

−6 −4 −2 0 2 4 6

Magnetic-reference contrast, $d_{NM} - d_{mag}$

C Nearest magnetic reference branch

Fraction among magnetic-like candidates

0.0 0.2 0.4 0.6 0.8 1.0

MatterGen base
MatterGen tuned
Con-CDVAE base
Con-CDVAE tuned
Crys-JEPA base
Crys-JEPA refined
CrystalFormer base
CrystalFormer tuned
CrystALLM base
CrystALLM tuned

D Retrospective DFT assessment

MatterGen base
Con-CDVAE base
CrystalFormer base
CrystALLM base

0.0 0.2 0.4 0.6 0.8 1.0

Fraction with $|m_{max}| > 0.5\ \mu_B$ in DFT